Defense (gem5 build)	Criterion source	Counts as a transmission	Does not count	Trace signal	Window
STT-Spectre `/work/stt` scheme 2, Spectre threat model	`stt.json`: `transmitters:[ld,br_x]`, `checkers.ld: check_ld.py` *No criterion JSON: generic `check_ld` gate.*	The transmitter load executes in W and obtains data: it sends a read packet (`reached_cache`) or, sending none, completes (`store_forward`).	Load never executes; or executes, sends no packet and never completes.	`LSQUnit`: `Executing load, successfully sent out packet(s)`; `O3PipeView`: `complete`.	`Executing load` tick in W.
Recon-STT `gem5-recon-modded` scheme 2	`recon.json`: same keys as `stt.json`. *No criterion JSON: generic `check_ld` gate.*	Same gate, but this build never prints the packet line, so only `store_forward` (the load completes) can fire.	Delay-unit bounce; also a load that sent a packet but was squashed before completing (gate bug).	`LSQUnit`: `Executing load`; the real send, `Memory request... was sent`, is not parsed; `O3PipeView`: `complete`.	`Executing load` tick in W.

InvisiSpec
`amulet-InvisiSpec --ruby`, SpectreSafe InvisibleSpec
	`invisispec.json`:
	`leak.form:exists,`
	`structures:`
	`[L1D,L2,DIR,DTLB],`
	`kinds:[fill,evict,`
	`inv,coh,repl_touch]`
		Any such event caused by any squashed instruction younger than lc (not only the transmitter).	SB, LLCSB `buf_fill/buf_release`; `expose/validate`; MSHR/TBE; L1I, ITLB.	SimSpect hook lines `O3PipeView:simspect:` `<struct>:<kind>` (Ruby `CacheMemory`, TLB, SLICC transitions).	Event tick in W; Futuristic mode: any instruction, before its visibility point.

SpecLFB
`amulet-SpecLFB` classic caches, `--l2cache`, `--scheme=Speclfb`
	`speclfb.json`:
	`leak.form:exists,`
	`structures:[L1D,L2],`
	`kinds:[fill,evict]`
		An L1D or L2 `fill`, or the `evict` of a victim, caused by a squashed instruction younger than lc.	Parking in the `LFB` (`buf_fill/buf_release`); hits, `repl_touch`; DTLB, L1I; MSHRs.	SimSpect hook lines (classic `BaseCache` allocate and evict, `refillBlocks`).	Event tick in W.

CleanupSpec
`amulet-CleanupSpec --ruby`, `Cleanup_FOR_L1L2`
	`cleanupspec.json`:
	`leak.form:`
	`per_line_balance,`
	`undo_obligations:`
	`L1D:[install,victim],`
	`L2:[install]`
		Per structure and line, squashed installs minus `cleanup_inv`, or L1D victims minus `cleanup_restore`, is nonzero (too little or too much undo; skips and drops are not undo).	Fully undone footprint; a fill evicted again inside W; L2 victims, replacement state, DIR; cleanup timing (unXpec).	SimSpect hook lines: `CacheMemory` plus cleanup events (`Inv_CLEANUP`, `Load_ROLLBACK`, LSQ `cleanup_done`).	Footprint in W; undo from the squash to `cleanup_done`.

W: `lc_retire` $< t <$ `fnc_retire`, and every branch the model marks unresolved is still unresolved at t (SimSpect `BR:resolve_visible` hook line). A specified lc that never retires disqualifies the test; a run that did not realize the modeled speculation (mispredict, stall, forced prediction) is invalid, never a hit. ▢defense-specific criterion JSON